

Machine Learning for social sciences

Natural Language Processing : Session 1

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As you already know, machine learning can be used for many use cases :

- Value prediction
- Multi-label classification
- Similarity measurement
- and even more...

But what if we want to regroup data into groups without supervising the definition of these groups ?

Supervised Learning

- Learns from labeled data to make predictions or classify data.
- **Key Characteristics :**
 - Data includes input-output pairs.
 - Goal : Minimize error by matching predictions to known outputs.

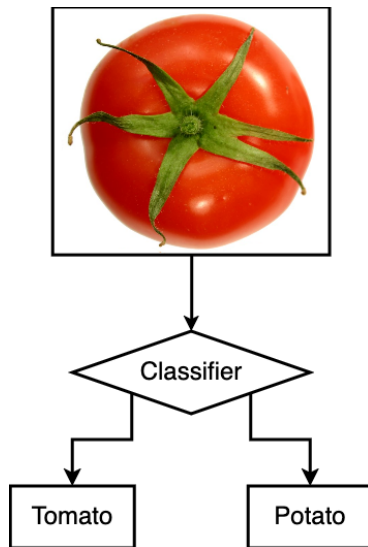
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 - Image classification
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Supervised vs. unsupervised learning

Supervised Learning

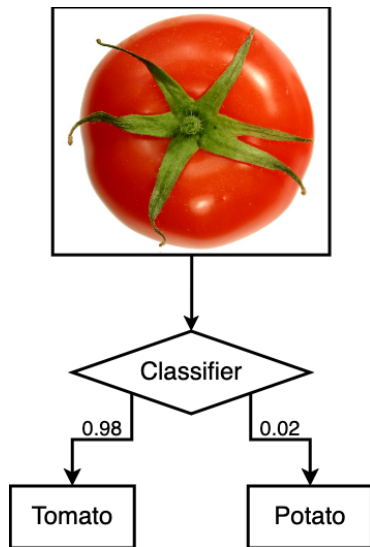
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- **Examples :**
 - Customer segmentation
 - Dimensionality reduction
 - Anomaly detection

Unsupervised Learning : the example of semantic textual similarity

Objective : Quantifying the degree of similarity between two texts based on their meaning rather than their vocabulary.

Supervised vs. unsupervised learning

Unsupervised Learning : the example of semantic textual similarity

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Example :

- **Sentence 1 :** This NLP class is fantastic 🥰
- **Sentence 2 :** I really like this natural language processing course
- **Similarity :** 0.703

. Results obtained using `sentence-transformers/all-mpnet-base-v2`

Supervised vs. unsupervised learning

Unsupervised Learning : the example of semantic textual similarity

Objective : Quantifying the degree of similarity between two texts based on their meaning rather than their vocabulary.

Example :

- **Sentence 1 :** This NLP class is fantastic 🥰
- **Sentence 2 :** This class is extremely boring 😴
- **Similarity :** 0.491

Supervised vs. unsupervised learning

Unsupervised Learning : the example of semantic textual similarity

Objective : Quantifying the degree of similarity between two texts based on their meaning rather than their vocabulary.

Example :

- **Sentence 1 :** This NLP class is fantastic 🥰
- **Sentence 2 :** I would like a falafel sandwich with Algerian sauce and a drink please.
- **Similarity :** 0.081

. Results obtained using `sentence-transformers/all-mpnet-base-v2`

Clustering is an unsupervised learning technique used to group similar data points. It identifies patterns or structures in the data without requiring labels.

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- K-Means Clustering
- Louvain Clustering
- DBSCAN (Density-Based Spatial Clustering)

A clustering algorithm : K-Means

① Initialization

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3 Update Step

- Recalculate each cluster's centroid as the mean of all points assigned to that cluster.

$$\mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i$$

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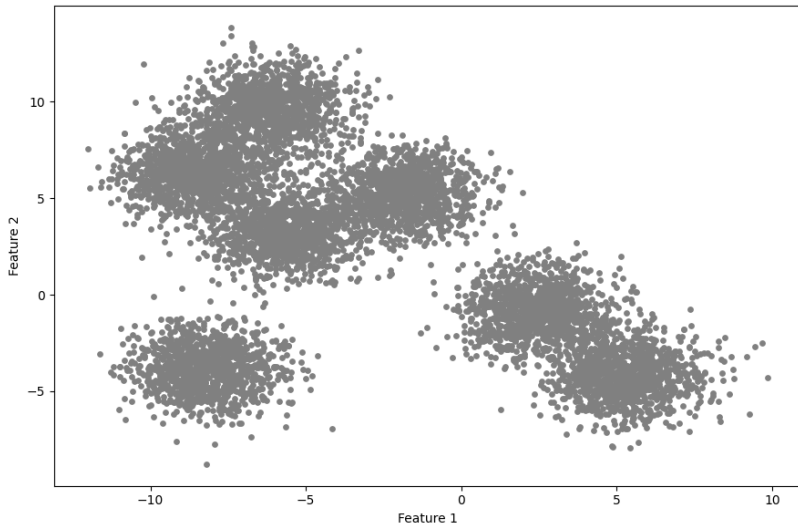
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4 Convergence Check

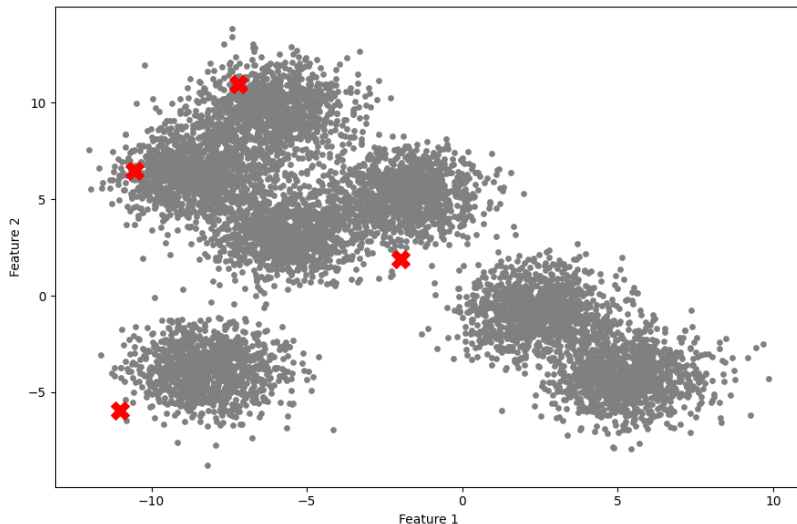
- Repeat the assignment and update steps until :
 - Centroids no longer change significantly, or
 - Cluster assignments stabilize, or
 - A predefined iteration limit is reached.

K-Means illustration



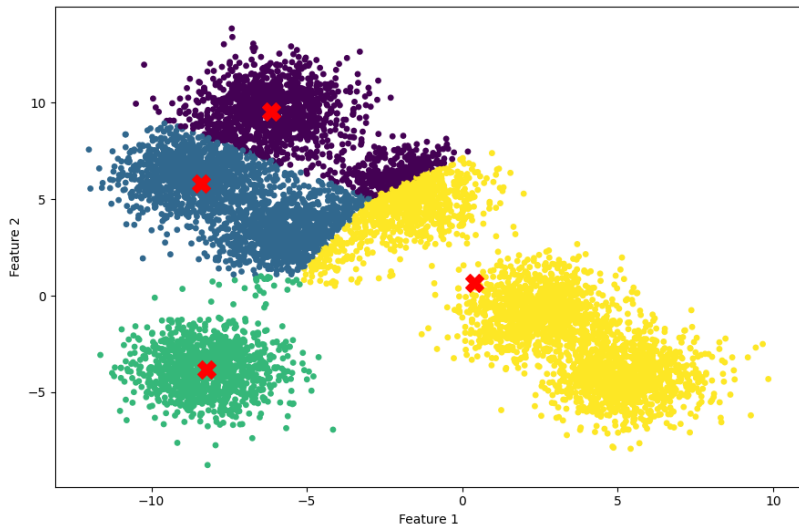
Base data

K-Means illustration



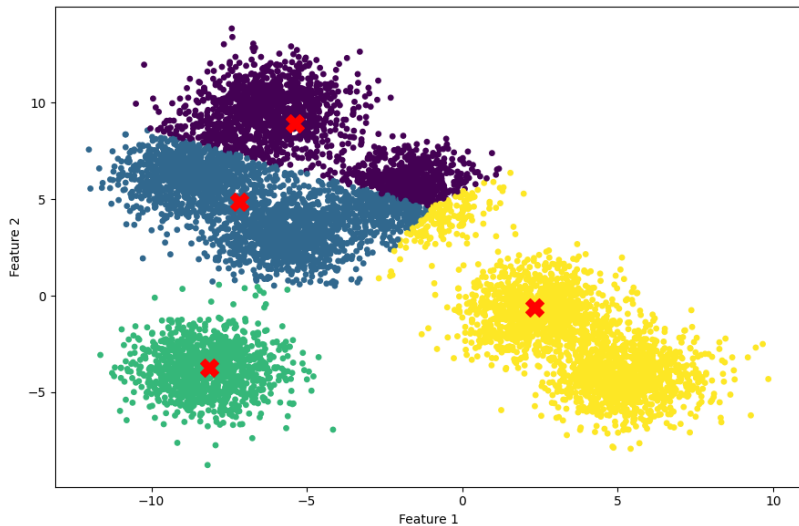
Initial random centroids (with $K = 4$)

K-Means illustration



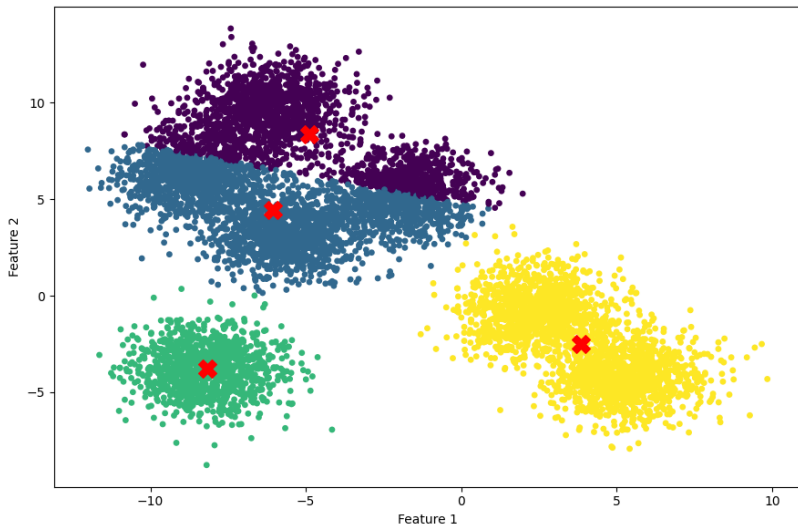
Iteration 1

K-Means illustration



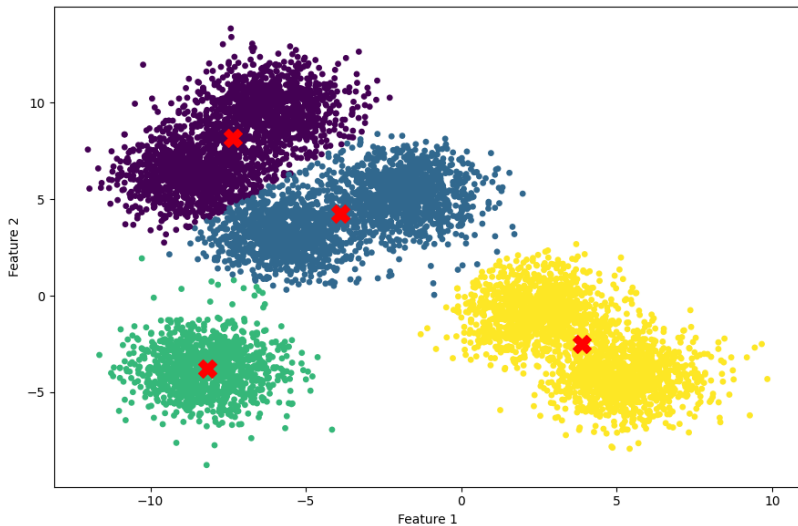
Iteration 2

K-Means illustration



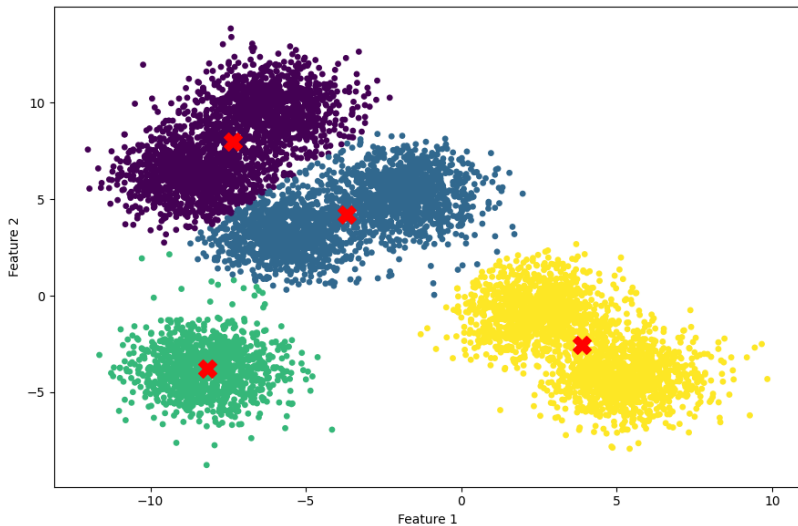
Iteration 5

K-Means illustration



Iteration 10

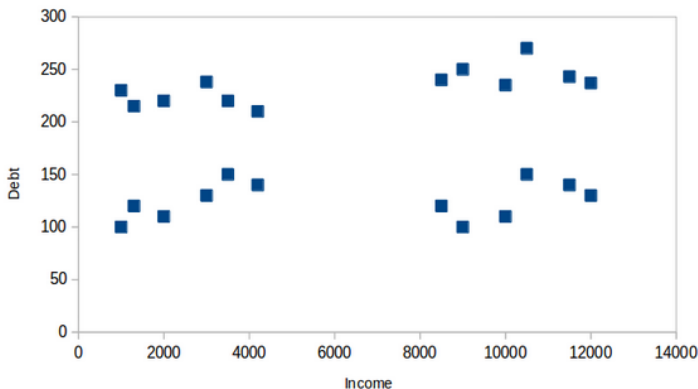
K-Means illustration



Iteration 50

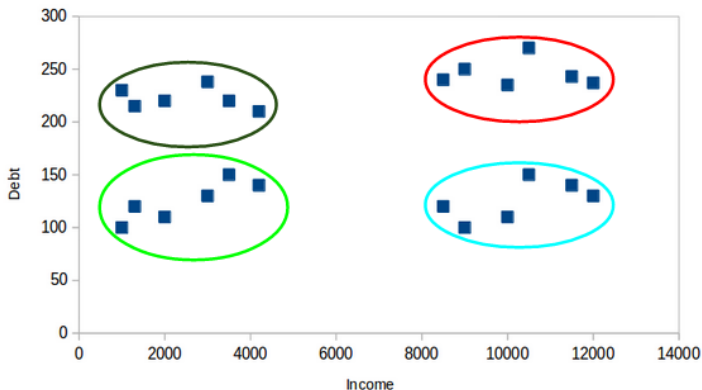
An example application of clustering

Example : K-Means clustering of people based on their debt and annual income (with $K = 4$)



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But we are in an NLP class, can we cluster texts ? Like, based on topics ? 🤔

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Applications :

- Document categorization
- Information retrieval
- Recommender systems
- Social media analysis

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LDA uncovers :

- Topic 1 (Sports) : *game, team, player, score, match.*
- Topic 2 (Politics) : *election, policy, government, vote.*
- Topic 3 (Technology) : *software, AI, computer, NLP.*

Lab 1 : Exploration of an online participatory process using LDA

Let's dive into it 🧐

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So how can we solve (some of) these issues ?

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Combines the strength of **transformer-based embeddings** (e.g., BERT) with techniques like UMAP for dimensionality reduction.

How does it work ?

- **Text Embedding** : Text is converted into embeddings (i.e., vectors) using models like BERT or SBERT.

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- **Text Embedding** : Text is converted into embeddings (i.e., vectors) using models like BERT or SBERT.
- **Dimensionality Reduction** : UMAP reduces embedding dimensions for efficient clustering.
- **Clustering** : HDBSCAN groups similar data points into clusters.
- **Topic Representation** : Key terms are extracted from clusters to represent topics.

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Let's see what we can do with it 🤓